

AI ENGINEER

CAREER PATH

PROGRAM

6-Month Intensive Schedule

From Beginner Developer to Production-Ready AI Engineer

3×

Sessions
Per Week

26

Total
Weeks

78

Sessions
Total

15+

Modules
Covered

INSTRUCTOR: PAUL OLAGBOYE

What You'll Learn



Software Engineering

Python, APIs, Git, Web Dev Fundamentals



AI & ML Foundations

LLMs, Data Handling, Visualization



LLM Engineering

Prompt Engineering, API Integration, SDKs



RAG & Knowledge

Embeddings, Vector DBs, Retrieval Pipelines



AI Agents

Orchestration, Memory, Multi-Agent Systems



Deployment & Security

Docker, Cloud, CI/CD, Responsible AI

8-Phase Learning Roadmap

PH 01

**Software Engineering
Foundations**

Modules 1–2

PH 02

AI & ML Foundations

Modules 3–5

PH 03

**LLM Engineering & AI
Applications**

Modules 6–7

PH 04

**Retrieval & Knowledge
Systems**

Modules 8–9

PH 05

AI Agents & Orchestration

Modules 10–12

PH 06

**AI Infrastructure &
Deployment**

Modules 13–14

PH 07

AI Security & Governance

Module 15

PH 08

**Career Specialization +
Capstone**

Final Phase

6 - MONTH SCHEDULE AT A GLANCE

26 Weeks · 3 Sessions Per Week · 78 Total Sessions

Month 1

Weeks 1–4 · Phase 1–2

Engineering & AI Foundations

12 sessions

Month 2

Weeks 5–8 · Phase 3

LLM Engineering & Prompt Mastery

12 sessions

Month 3

Weeks 9–13 · Phase 4

RAG & Knowledge Systems

15 sessions

Month 4

Weeks 14–18 · Phase 5

AI Agents & Orchestration

15 sessions

Month 5

Weeks 19–22 · Phase 6–7

Infrastructure, Deployment & Security

12 sessions

Month 6

Weeks 23–26 · Phase 8

Career Tracks & Capstone

12 sessions

Engineering & AI Foundations

Week 1

Module 01 – Programming Fundamentals

- Mon — Python basics, variables, data types, control flow
- Wed — Functions, OOP, classes & Git basics
- Fri — CLI tools, file handling, error handling — Lab: CLI Calculator

Week 2

Module 01–02 – Python & APIs

- Mon — Modules, packages & project structure
- Wed — HTTP, REST APIs, JSON — Weather API App
- Fri — Auth basics, async programming — Lab: Auth System

Week 3

Module 02–03 – Web Dev & Intro to AI

- Mon — Database fundamentals — REST Dashboard
- Wed — ML vs Deep Learning, Generative AI, LLMs
- Fri — AI use cases, risks & responsible AI — Prompt Playground

Week 4

Module 04–05 – Data Handling & Visualization

- Mon — NumPy & Pandas — data cleaning & transformation
- Wed — Feature engineering, pipelines — Dataset Cleaner
- Fri — Matplotlib, Seaborn, EDA — AI Analytics Dashboard

LLM Engineering & Prompt Mastery

Week 5

Module 06 – Prompt Engineering

- Mon — Zero-shot & few-shot prompting fundamentals
- Wed — Chain-of-thought, prompt templates & patterns
- Fri — Structured outputs, optimization — AI Writing Assistant

Week 6

Module 06 – Evals-Driven Development

- Mon — RAGAS for RAG evaluation & LLM-as-judge patterns
- Wed — Prompt → eval → iterate workflow
- Fri — Regression testing AI behaviour — Prompt Library System

Week 7

Module 07 – AI APIs & SDK Integration

- Mon — OpenAI, Anthropic, HuggingFace APIs
- Wed — Streaming, function calling, token management
- Fri — Assistants API — Lab: AI Chatbot

Week 8

Module 07 – Structured Outputs & Validation

- Mon — Pydantic validation pipelines
- Wed — Instructor & Outlines libraries, JSON mode & schema enforcement
- Fri — Output quality guardrails — Resume Analyzer & Code Explainer

RAG & Knowledge Systems

Week 9

Module 08 – Embeddings & Vector DBs

- Mon — Semantic search & embedding models
- Wed — Chunking strategies & metadata handling
- Fri — Pinecone, ChromaDB, pgvector — AI Knowledge Base

Week 10

Module 08–09 – RAG Foundations

- Mon — RAG architecture & pipeline design
- Wed — Context injection & citation systems
- Fri — Hallucination reduction — Lab: PDF Chat

Week 11

Module 09 – AI Observability & Tracing

- Mon — LangSmith, Arize, Helicone setup
- Wed — Token-level tracing & latency monitoring
- Fri — Hallucination rate & cost dashboards per feature

Week 12

Hybrid Search & Reranking

- Mon — BM25 + semantic hybrid search
- Wed — Cohere Rerank & cross-encoders
- Fri — Neo4j graph DBs & advanced chunking — Semantic Search Engine

RAG & Knowledge Systems — Advanced

Week 13

Module 09 – Advanced RAG & Enterprise Knowledge

- Mon — Enterprise RAG patterns & multi-document retrieval
- Wed — AI Research Assistant build & evaluation
- Fri — Enterprise Knowledge Bot — Phase 4 review & portfolio update

AI Agents & Orchestration

Week 14

Module 10 – AI Agents & Tool Use

- Mon — Agent architectures — LangChain & LangGraph intro
- Wed — CrewAI & MCP fundamentals
- Fri — Tool calling, function execution — AI Research Agent

Week 15

Module 10 – Browser & Workflow Agents

- Mon — Browser automation agents
- Wed — Multi-step task completion patterns
- Fri — Lab: AI Workflow Assistant

Week 16

Module 11 – Context Engineering & Memory

- Mon — Long-term memory architecture design
- Wed — State management & context window optimization
- Fri — Session-aware token-efficient systems — Persistent AI Workspace

Week 17

Module 11–12 – Memory & Multi-Agent

- Mon — Token-efficient context engineering
- Wed — Multi-agent orchestration — task delegation
- Fri — Reporting loops, workflow isolation — AI Project Manager

AI Agents — Multi-Agent Systems

Week 18

Module 12 – Multi-Agent Systems at Scale

- Mon — Agent coordination at scale — orchestration patterns
- Wed — AI Dev Team Simulator & AI QA + Dev Workflow
- Fri — Multi-Agent Task Platform — Phase 5 review & portfolio update

Infrastructure, Deployment & Security

Week 19

Module 13 – Backend & Cloud

- Mon — FastAPI / Node.js backends for AI services
- Wed — Redis, queues & async workers
- Fri — WebSockets & server-sent events

Week 20

Module 13–14 – Docker, Cloud & Caching

- Mon — Docker & containerization
- Wed — AWS / GCP / Azure deployment
- Fri — Semantic caching (GPTCache), prompt caching, LiteLLM model routing

Week 21

Module 14 – CI/CD, Monitoring & Optimization

- Mon — CI/CD pipelines for AI systems
- Wed — Monitoring, logging, scaling & cost optimization
- Fri — Production AI SaaS Deployment — Cloud-hosted AI Assistant

Week 22

Module 15 – AI Security & Governance

- Mon — Prompt injection, jailbreak protection, sandboxing AI outputs
- Wed — Data privacy, PII detection & redaction, access control
- Fri — Guardrails AI, NeMo Guardrails — Secure AI Assistant & Moderation System

Career Tracks & Capstone

Week 23

Track Module – Specialization

- Mon — Choose track: Fullstack / Backend / Product / Automation / Systems
- Wed — Track-specific deep dive — advanced patterns & gaps
- Fri — Capstone scoping & proposal submission

Week 24

Capstone – Build Sprint 1

- Mon — Architecture design & tech stack setup
- Wed — Core feature implementation — mentor check-in
- Fri — Integration, testing & observability setup

Week 25

Capstone – Build Sprint 2

- Mon — Deployment & production hardening
- Wed — Evals, quality checks & performance tuning
- Fri — Portfolio documentation, README & case study write-up

Week 26

Demo Day & Career Readiness

- Mon — Final capstone polish & peer review
- Wed — Demo day — present to cohort & instructors
- Fri — Career readiness: portfolio review, LinkedIn, job search strategy

5 Career Specialization Tracks

AI Fullstack

Phases 1–7 cover this well

TIER 2 PRIORITY

MODERATE

Multimodal · Evals-driven dev

CAPSTONE OPTIONS

AI SaaS Dashboard · AI
Content Platform · AI Booking
Assistant

AI Backend

Phase 6 covers this well

TIER 2 PRIORITY

SIGNIFICANT

LiteLLM routing · vLLM local
deploy · Evals workflow

CAPSTONE OPTIONS

High-throughput AI API · Multi-
model inference gateway

AI Product

Partial: product skills gap

TIER 2 PRIORITY

LOW

Legal & AI Act basics · Evals-
driven dev

CAPSTONE OPTIONS

AI feature spec + prototype +
eval report

AI Automation

Phase 5 covers this well

TIER 2 PRIORITY

MODERATE

Multimodal pipelines · Evals-
driven

CAPSTONE OPTIONS

Multi-agent business workflow ·
AI document pipeline

AI Systems

Partial: reliability gap

TIER 2 PRIORITY

CRITICAL ★

LoRA/QLoRA fine-tuning ·
vLLM · LiteLLM

CAPSTONE OPTIONS

Production AI infra with SLO
monitoring

Technologies & Capstone Projects

LANGUAGES

Python · JavaScript / TypeScript

AI FRAMEWORKS

LangChain · LangGraph · CrewAI ·
HuggingFace

DATABASES

PostgreSQL · Redis · pgvector · Pinecone ·
ChromaDB

DEPLOYMENT

Docker · AWS · GCP · CI/CD Pipelines

AI APIS

OpenAI · Anthropic · HuggingFace

OBSERVABILITY

LangSmith · Arize · Helicone · GPTCache

Graduate Roles:

AI Engineer · LLM Engineer · AI Fullstack Dev · AI Backend Engineer · AI Automation Engineer · AI Systems Engineer

Capstone Project Options

AI Fraud Detection Platform

AI Recruitment Assistant

AI Coding Workspace

AI Customer Support System

AI Research Platform

Multi-Agent Workflow System

PROGRAM
PHILOSOPHY

**AI should not replace
engineering fundamentals.**

AI should amplify
engineering capability.

Students are trained not only to use AI tools, but to engineer scalable, maintainable, secure, and production-ready AI-native systems.

**6
Months**

Intensive

3×/Wk

Sessions

78

Sessions

5

Tracks

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